
Yang LIN

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Professional Objective

Generative AI engineer who works great unsupervised, and so do my models. Trained LLMs from scratch, cut diffusion sampling from 50 steps to 1, and ready for the next hard problem.

Education

02/2025 - Present **The Australian National University** **Expected Grad:** Nov 2026
Canberra, Australia Master of Computing (Specialisation in Artificial Intelligence & Machine Learning)

02/2022-11/2024 **The Australian National University** **GPA:** 6.087/7.0
Canberra, Australia Bachelor of Applied Data Analytics (Major in Statistics)

Work Experience

07/2023-11/2023 **Papua New Guinea Institute of Medical Research** | Intern | Database Development
Remote | Canberra

- Designed and maintained a database supporting **50+ research projects**, managing multi-dimensional data with a focus on data integrity and consistency.
- Built automated **ETL pipelines using Python and SQL**, reducing manual processing time by **75%** while maintaining **99%+ data accuracy**.
- Trained **15+ users** and developed documentation to support consistent data management and long-term data governance.

Skills summary

- Languages & Tools:** Python, R, SQL, Java; Power BI, NumPy, Pandas, Linux, Git, Docker.
- ML&RL:** Regression, decision trees, SVM, hyperparameter tuning; MDP/POMDP, policy iteration, Q-learning.
- LLM & Gen-AI:** Transformers, LLaMA, Qwen, SFT, LoRA, prompt engineering, RAG, LangChain agents.
- AI & Planning:** search algorithms, AI agents, motion planning, EDA, statistical modeling, data visualisation.

Projects

GPT-style Language Model trained from scratch (nanoGPT) 03/2026 – 04/2026

Tech: Python, PyTorch, Transformers, HuggingFace

- Built an **end-to-end training pipeline** for the ROCStories corpus, implementing **GPT-2 BPE** tokenization, context packing, and dataset preprocessing to maximise context utilization.
- Designed and trained a **31.7M**-parameter Transformer language model from scratch, incorporating SwiGLU MLP and RMSNorm; tuned learning rate, warmup, weight decay to achieve **15.2 test perplexity** and **1st place** in an LLM story generation competition.

Diffusion model: reproduction & study Flow Matching & MeanFlow 04/2026 – 05/2026

Tech: Python, PyTorch, Flow Matching, MeanFlow

- Reproduced the Flow Matching training and sampling pipeline from *Back to Basics*; benchmarked **x-, ϵ -, and v-prediction** parameterisations and validated results against the paper.
- Investigated the degradation of v-prediction in high-dimensional spaces through controlled experiments; identified the **x-prediction with v-loss** as the most effective configuration under high-dimensional settings.
- Implemented MeanFlow for single-step generation by modelling the interval-averaged velocity field, reducing ODE sampling from multi-step inference to **NFE = 1** while maintaining competitive generation quality.

Multi-Drone Motion Planning & Decision-Making under uncertainty 08/2025 – 09/2025

Tech: Python, Reinforcement Learning, MDPs/POMDPs

- Designed centralized motion planning and collision-free navigation algorithms for multi-drones in complex dynamic environments; achieved **0% collision rate** for 5 drones across 8 obstacle scenarios, with an average **planning time of 949 ± 424 ms**.
- Formulated action uncertainty and wind disturbances as an MDP; solved the optimal policy via policy iteration, achieving a task **success rate of $83.3 \pm 14.2\%$** under perturbations in single-drone, 2-obstacle scenarios.